

Residual-shape inference and external-proxy audit of structural disturbance in SPARC rotation curves

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Abstract

We ask whether fixed rotation-curve residual-shape features can recover externally reviewed structural-disturbance labels in SPARC galaxies, and whether independent HI disturbance proxies support the same diagnostic direction. The study reverses the Paper 1 audit: the A/C labels are treated as frozen external targets, while residual features are evaluated as predictors under leave-one-galaxy-out thresholding, shuffled-label null tests, bootstrap uncertainty, baseline-family controls, and observability stress checks. The primary internal feature, Projection RMS, reaches LOOGO AUC=0.771008403 with shuffled-label $p = 0.002000000$ and bootstrap 95% AUC interval [0.600802469, 0.909100262]. MOND-simple and empirical RAR-like residual scores also separate A/C systems, whereas a Newtonian baryonic RMS control is near chance, indicating a low-acceleration residual-family effect rather than projection-formula uniqueness. External proxy readouts are supportive but not decisive. The result is therefore a reproducible residual-inference and external-proxy audit, not a Tau Core validation claim, not gravity-model selection, and not independent paper-grade external validation.

1 Introduction

Rotation-curve residuals are usually discussed as model error, but their radial structure can also carry information about non-equilibrium dynamics, non-circular motions, pressure support, beam smearing, inclination, and source-dependent observability [8, 6]. Paper 1 established a residual-blind association between externally assigned structural-disturbance labels and low-acceleration residual scatter in SPARC. Here we ask the inverse diagnostic question: can fixed residual-shape features recover those external labels better than chance?

The scope is deliberately narrow. This paper does not use residuals to redefine the labels, does not validate Tau Core, and does not select a unique gravity law. It tests whether a frozen residual feature map contains recoverable information about externally reviewed A/C disturbance classes, then checks whether independent HI disturbance proxies point in the same direction.

2 Data and frozen inputs

The working A/C sample contains 45 SPARC galaxies inherited from the Paper 1 reproducibility packet: 17 externally regular A systems and 28 externally disturbed C systems. SPARC rotation-curve and mass-model context follows [3, 2]. B-class systems remain excluded from the primary target labels because they encode uncertainty by construction. The residual features are computed from the fixed Paper 1 point map and are not retrained on the external-proxy catalogues.

The public packet contains the derived tables, scripts, figures, and audit records needed to regenerate the diagnostic results. Raw survey products and raw SPARC rotmod files are not redistributed by this repository.

3 Residual-shape endpoint and validation design

The primary endpoint is Projection RMS, the galaxy-level RMS of a fixed low-acceleration projection residual. It is used as an operational residual-shape score rather than as a physical proof of the projection formula. The key design choice is that the score is fixed before the external-proxy readouts are interpreted.

The primary internal validation uses leave-one-galaxy-out thresholding. For each held-out galaxy, the decision threshold is recomputed from the remaining A and C systems as the midpoint between their class medians. A shuffled-label null preserves class counts and tests whether the observed AUC is expected under random A/C labels. Bootstrap resampling estimates sample-size uncertainty. These tests are internal to SPARC and should not be described as independent validation.

4 Primary results

Table 1: Primary residual-inference results.

Quantity	Value	Interpretation	Claim boundary
Projection RMS LOOGO AUC	0.771008403	primary diagnostic	SPARC-internal
Projection RMS shuffled-label p	0.002000000	label-null check	not replication
MOND-simple RMS AUC	0.720588235	low-acceleration family	not unique
RAR-like RMS AUC	0.731092437	low-acceleration family	not model selection
Newtonian baryonic RMS AUC	0.506302521	near-chance control	limited inference

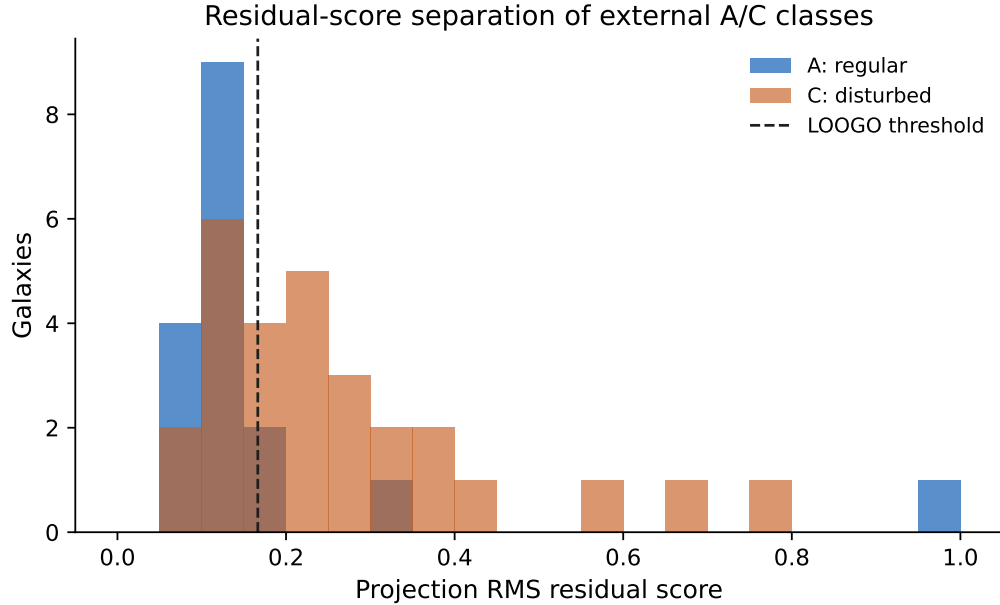


Figure 1: Distribution of the fixed Projection RMS residual score for externally regular A and disturbed C systems.

The primary signal is positive and statistically sharp in the frozen internal packet. The result is strongest when stated as class recovery from residual shape, not as a physical detection of an underlying field.

5 Baseline-family comparison

The baseline comparison weakens any projection-specific claim but strengthens the broader phenomenological result. MOND-simple [5] and empirical RAR-like [4] residual scores also separate A/C systems, while the Newtonian baryonic RMS score is near chance. The defensible conclusion is that A/C separation is concentrated in low-acceleration residual-family scores.

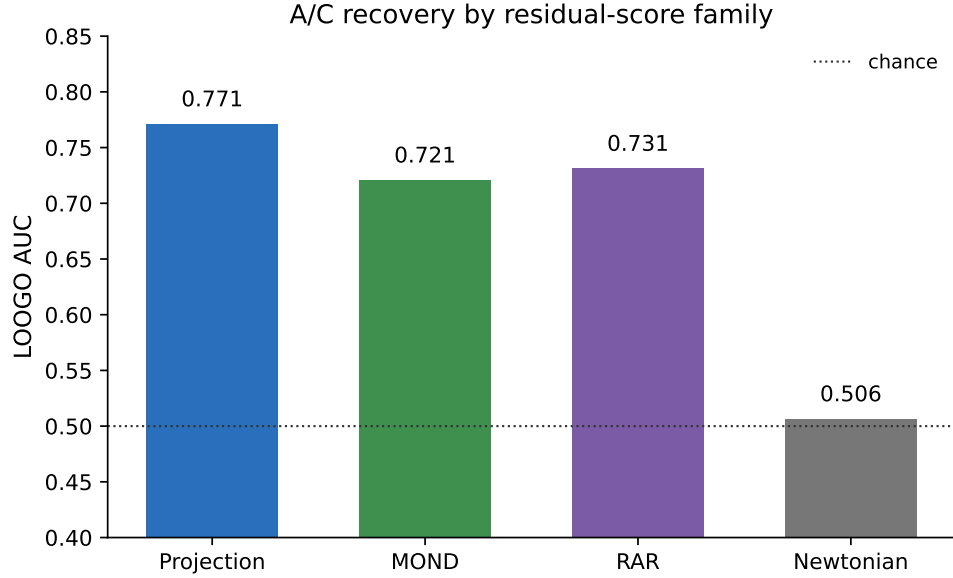


Figure 2: Held-out A/C recovery across residual-score families. The Newtonian baryonic RMS score is near chance.

This pattern is compatible with several physical readings: disturbed systems may violate smooth equilibrium assumptions, non-circular motions may be more important in low-acceleration regions, or observability may expose more disturbance structure in particular subsets. The present data do not distinguish these explanations.

6 Failure modes

The error audit is retained because a useful diagnostic must show where it fails. False-negative C systems demonstrate that externally disturbed galaxies do not always produce large residual burden under a smooth rotation-curve score. False-positive A systems show that residual structure can arise without a strong external disturbance label.

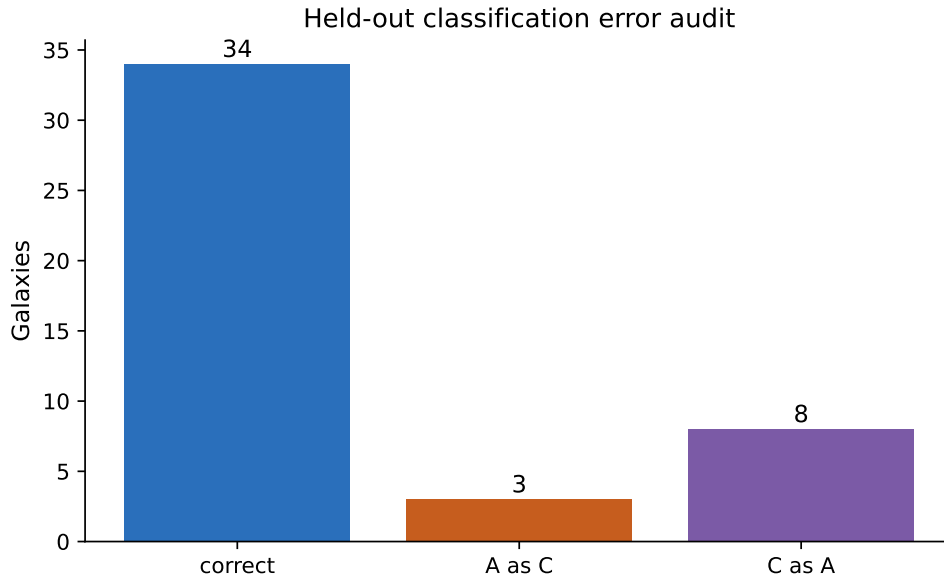


Figure 3: Held-out classification outcome counts for the Projection RMS threshold rule.

These failures are not relabeling evidence. They are targets for follow-up inspection, especially where residual morphology, inclination, radial coverage, and HI kinematic asymmetry disagree.

7 External proxy readouts

The external HI context draws on WHISP lopsidedness and morphology [9, 10, 1], THINGS non-circular-motion diagnostics [8], Reynolds et al. HI asymmetry catalogues [7], and ALFALFA profile asymmetry [11]. WHISP resolved-HI asymmetry is the strongest supportive external readout, but its overlap is small. WHISP morphology is mixed, Reynolds/LVHIS is promising but below the frozen $N \geq 15$ gate, and ALFALFA/HALOGAS do not provide strong directional support. The external evidence therefore supports the paper as an audit, not as an independent validation result.

8 THINGS route2 negative audit

The THINGS expansion route was audited because it could have raised the independent-control overlap. It is now closed as not score-ready. The required machine-readable per-radius mass-model columns were not recovered for the missing galaxies, and the reconstruction route failed its photometry and solver-validation gates before any missing-galaxy scores were computed.

This negative audit should remain visible. It documents why the paper does not claim THINGS $N \geq 15$, avoids plot-digitized or synthetic mass models, and prevents endpoint-driven data recovery from entering the evidence chain.

9 Observability and systematics

Observability remains the main scientific caveat. Nearby galaxies can reveal asymmetry and morphological disturbance more easily than distant systems; inclination, beam smearing, asymmetric

drift, pressure support, and non-circular motion can all alter residual morphology. The current controls reduce but do not eliminate these concerns.

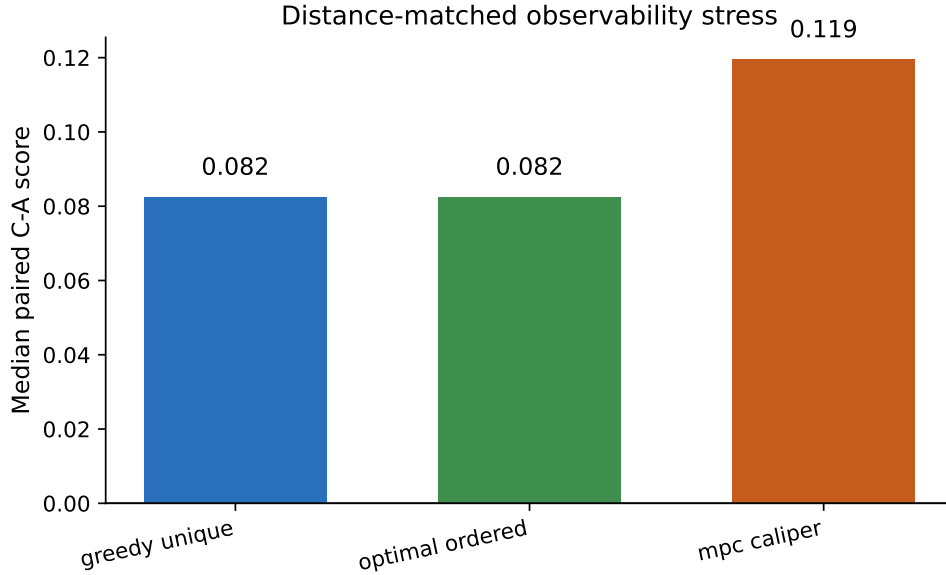


Figure 4: Distance-matched stress checks for the primary residual score. These checks reduce but do not eliminate observability concerns.

Accordingly, the result should be read as a reproducible SPARC diagnostic association with external-proxy context, not as a selection-function-proof physical inference.

10 Claim boundary and Phase II

Allowed claim: fixed residual-shape features recover externally reviewed A/C disturbance class better than chance in the current SPARC packet, and several external proxy readouts provide mixed but informative context.

Forbidden claims: Tau Core validation, gravity-model selection, projection-formula uniqueness, replacement of external labels by residual-only labels, broad independent external validation, or THINGS route2 positive evidence.

The next paper-grade step is a held-out external source-family test with $N \geq 15$, a frozen evidence rule, no velocity-endpoint refit, explicit observability covariates, and predefined failure conditions. A negative Phase II result should be treated as evidence that the present association is SPARC-specific, proxy-specific, or observability-driven.

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